Digital Image Processing (PCD)

Image Segmentation and Morphological Processing
(Assignment 03)

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I. Introduction

Image segmentation is a technique in digital image processing that divides an image into meaningful regions or objects. In this assignment, thresholding is used to separate the object from the background based on pixel intensity values. After segmentation, four morphological operations, which are erosion, dilation, opening, and closing, are applied to see how they affect the segmentation result. All methods in this assignment are implemented from scratch without using image processing libraries.

II. Image Used

The image used in this assignment is a photo of a strawberry on a white background, resized to 512×512 pixels. This image was chosen because there is a clear difference between the dark red strawberry and the bright white background, making it suitable for thresholding. The strawberry also has natural details, such as seeds and surface textures, which can create small holes and noise in the segmentation result. This makes it suitable for testing the effects of morphological operations.

III. Segmentation Method (Thresholding)

Thresholding is implemented from scratch by checking every pixel in the grayscale image and comparing its value with a threshold of 150. If a pixel value is below 150, it is assigned white (255) to represent the object. If its value is equal to or above 150, it is assigned black (0) to represent the background. The result shows the strawberry as a white region on a black background. However, due to the natural texture of the strawberry, such as its seeds and shiny spots, the result contains small black holes and noise inside the white region.

IV. Morphological Methods

All morphological operations use a 5×5 kernel and are implemented from scratch using manual pixel loops.

4.1 Erosion

Erosion works by checking each pixel and its surrounding 5×5 region. A pixel remains white only if all pixels in that region are also white. Otherwise, it is changed to black. The result

shows the strawberry becoming smaller and thinner. Thin parts, such as the stem and leaves, mostly disappear. The holes inside the fruit also become more visible because erosion removes white pixels from the edges inward.

4.2 Dilation

Dilation works by checking each pixel and its surrounding 5×5 region. A pixel becomes white if at least one pixel in that region is white. The result shows the strawberry becoming larger, while most of the black holes inside the fruit are filled in. This creates a more solid white region and makes the overall shape expand compared to the original segmentation.

4.3 Opening

Opening is performed by applying erosion first, followed by dilation. The result shows some small noise being removed, while the overall shape becomes slightly smaller compared to the original segmentation. However, because the strawberry has many small details and textures, the erosion step also removes some parts of the fruit. This makes the result appear slightly fragmented and blocky.

4.4 Closing

Closing is performed by applying dilation first, followed by erosion. The result shows most of the black holes inside the strawberry being filled in, while the overall size and shape remain close to the original segmentation. This makes closing the most effective operation for this image because it produces a cleaner and more solid result without significantly changing the shape of the strawberry.

V. Analysis and Comparison

Method	**Effect on Shape**	**Effect on Holes**	**Best Use For**
Thresholding Only	Original shape	Small holes and noise remain visible	Basic segmentation
Erosion	Shape becomes smaller	Holes become more visible and thin parts disappear	Removing thin connections
Dilation	Shape becomes larger	Holes are filled and the object becomes more solid	Filling gaps and holes
Opening	Shape may become slightly smaller	Removes some small noise	Removing small noise
Closing	Shape remains similar	Most internal holes are filled	Filling internal holes

Table 5.1 Comparison of Segmentation Results With and Without Morphological Processing

VI. Conclusion

This assignment showed how thresholding can be used to separate an object from its background and how morphological operations can improve the segmentation result. Each operation produces a different effect on the shape and details of the object. Erosion and opening remove some parts of the object, while dilation and closing expand the object and fill small holes. Based on the results, closing was the most suitable operation for the strawberry image because it filled most of the internal holes while maintaining its overall shape. Overall, this assignment shows that choosing the right morphological operation depends on the condition of the segmented image and the result that we want.